

Translated

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Depth First Search

Depth First Search is one of the main graph algorithms.

Depth First Search finds the lexicographical first path in the graph from a source vertex u to each vertex. Depth First Search will also find the shortest paths in a tree (because there only exists one simple path), but on general graphs this is not the case.

The algorithm works in $O(m + n)$ time where n is the number of vertices and m is the number of edges.

Description of the algorithm

The idea behind DFS is to go as deep into the graph as possible, and backtrack once you are at a vertex without any unvisited adjacent vertices.

It is very easy to describe / implement the algorithm recursively: We start the search at one vertex. After visiting a vertex, we further perform a DFS for each adjacent vertex that we haven't visited before. This way we visit all vertices that are reachable from the starting vertex.

For more details check out the implementation.

Applications of Depth First Search

- Find any path in the graph from source vertex u to all vertices.
- Find lexicographical first path in the graph from source u to all vertices.
- Check if a vertex in a tree is an ancestor of some other vertex:

At the beginning and end of each search call we remember the entry and exit "time" of each vertex. Now you can find the answer for any pair of vertices (i, j) in $O(1)$: vertex i is an ancestor of vertex j if and only if $\text{entry}[i] < \text{entry}[j]$ and $\text{exit}[i] > \text{exit}[j]$.

- Find the lowest common ancestor (LCA) of two vertices.
- Topological sorting: